Go2 SDK Development Guide

Search article titles

About Go2
Remote Controller
SDK Concepts 3
Application Development 7
Software Interface Services 19
Motion Services Interface V2.0
DDS Services Interface
Avoidance Services Interface
VuiClient Interface
RobotStateClient Interface
Motion Switcher Service Interface
Sports Services Interface
AI motion Services interface
Basic Services Interface
LiDAR Services Interface
Multimedia Services Interface
UWB Services Interface
Fault Services Interface
SLAM and Navigation Services Interfa
Aruco Recharge Service Interface
Odometer Services Interface
D1 Mechanical Arm Services Interface
ROS2 Services Interface
Python Services Interface
Case reference 8
Debugging Specification
FAQ

Python Services Interface

Last Updated On : 2025-11-21 11:20:02

This page contains content

unitree_sdk2_python
Installation
Dependencies
Installing unitree_sdk2_python
FAQ
Usage
DDS Communication
High-Level State and Control
Low-Level State and Control
Wireless Controller Status Ac...
Front Camera
Obstacle Avoidance Switch
Light and Volume Control

unitree_sdk2_python

unitree_sdk2 python interface

Installation

Dependencies

- python>=3.8
- cyclonedds==0.10.2
- numpy
- opencv-python

Installing unitree_sdk2_python

Execute the following in the terminal:

```bash
cd ~
sudo apt install python3-pip
git clone [https://github.com/unitreerobotics/unitree_sdk2_python.git](https://github.com/unitreerobotics/unitre
cd unitree_sdk2_python
pip3 install -e .
```

FAQ

1. Error when running `pip3 install -e .`

```bash
Could not locate cyclonedds. Try to set CYCLONEDDS_HOME or CMAKE_PREFIX_PATH
```

This error indicates that the cyclonedds path cannot be found. First, compile and install cyclonedds:

```bash
cd ~
git clone [https://github.com/eclipse-cyclonedds/cyclonedds](https://github.com/eclipse-cyclonedds/cyclonedds) -
cd cyclonedds && mkdir build install && cd build
cmake .. -DCMAKE_INSTALL_PREFIX=../install -DBUILD_DDSPERF=OFF
cmake --build . --target install
```

Navigate to the `unitree_sdk2_python` directory, set `CYCLONEDDS_HOME` to the path where you just compiled cyclonedds, and then in stall `unitree_sdk2_python`:

```bash
cd ~/unitree_sdk2_python
export CYCLONEDDS_HOME="/{path to your cyclonedds}/cyclonedds/install"
pip3 install -e .
```

For details, see:
https://pypi.org/project/cyclonedds/#installing-with-pre-built-binaries

Usage

The python sdk2 interface remains consistent with the unitree_sdk2 interface, achieving robot status acquisition and control throug h request-response or publish-subscribe topics. Corresponding examples are located in the `/example` directory. Before running the examples, you need to configure the robot's network connection according to the document 《Quick Start》.

DDS Communication

Execute the following in the terminal:

```bash
python3 ./example/helloworld/publisher.py
```

Open a new terminal and execute:

```bash
python3 ./example/helloworld/subscriber.py
```

You should see the data information output in the terminal. The data transmitted by `publisher.py` and `subscriber.py` is defined i n `user_data.py`. Users can define the data structures they need to transmit as required.

Helpful
Feedback

High-Level State and Control

The data structures and control methods for the high-level interface are consistent with unitree_sdk2. For details, see: https://support.unitree.com/home/zh/developer/sports_services

High-Level Control

Execute the following in the terminal:

```bash
python3 ./example/go2/high_level/go2_sport_client.py [network_interface_name]
```

Where `[network_interface_name]` is the name of the network card connected to the robot. Please modify it according to the actual situation. Format examples: `enp2o`, `exp889dsxspod21`
This example provides several test methods; select one by inputting the sequence number to execute:

```python
TestOption(name="damp", id=0),
TestOption(name="stand_up", id=1),
TestOption(name="stand_down", id=2),
TestOption(name="move forward", id=3),
TestOption(name="move lateral", id=4),
TestOption(name="move rotate", id=5),
TestOption(name="stop_move", id=6),
TestOption(name="switch_gait", id=7),
TestOption(name="switch_gait", id=8),
TestOption(name="balanced stand", id=9),
TestOption(name="recovery", id=10),
TestOption(name="recovery", id=10),
TestOption(name="left flip", id=11),
TestOption(name="back flip", id=12),
TestOption(name="free walk", id=13),
TestOption(name="free bound", id=14),
TestOption(name="free avoid", id=15),
TestOption(name="walk stair", id=16),
TestOption(name="walk upright", id=17),
TestOption(name="cross step", id=18),
TestOption(name="free jump", id=19)
```

Low-Level State and Control

The data structures and control methods for the low-level interface are consistent with unitree_sdk2. For details, see: https://support.unitree.com/home/zh/developer/Basic_services

Low-Level Motor Control

First, use the app to turn off the high-level motion service (`sport_mode`), otherwise, it will cause command conflicts.
Execute the following in the terminal:

```bash
python3 ./example/go2/low_level/go2_stand_example.py [network_interface_name]
```

The rear-left thigh (hip) joint will be held at 0 degrees (for safety, $k_p = 10$ and $k_d = 1$ are set here), and the rear-left calf joint will continuously output $1\ \mathrm{Nm}$ of torque.

Wireless Controller Status Acquisition

Execute the following in the terminal:

```bash
python3 ./example/wireless_controller/wireless_controller.py [network_interface_name]
```

The terminal will output the status of each button. For the definition and data structure of the controller buttons, see: https://support.unitree.com/home/zh/developer/Get_remote_control_status

Front Camera

Use OpenCV to acquire the front camera (ensure the system has a graphical interface; press ESC to exit the program):

```bash
python3 ./example/go2/front_camera/camera_opencv.py [network_interface_name]
```

Obstacle Avoidance Switch

```bash
python3 ./example/obstacles_avoid_switch/obstacles_avoid_switch.py [network_interface_name]
```

The robot will loop through enabling and disabling the obstacle avoidance function. For details on the obstacle avoidance service, see https://support.unitree.com/home/zh/developer/ObstaclesAvoidClient

Light and Volume Control

```bash
python3 ./example/vui_client/vui_client_example.py [network_interface_name]
```

The robot will cycle through adjusting the volume and light brightness. For details on this interface, see https://support.unitree.com/home/zh/developer/VuiClient

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